solve all parts please describe all steps clearly an dfull steps to solve question all parts with full expanation expalin all maheatcalyy parts of each steps and formualas an dthoerem a best solution 21 to 36

Here are the step-by-step mathematical solutions for **Exercises 21 through 36**.

Core Theorem & Limit Rules Used

1. **Fundamental Trigonometric Limit:**

$$\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1$$

2. **Reciprocal Limit:**

$$\lim_{\theta \to 0} \frac{\theta}{\sin \theta} = 1$$

3. **Limit Substitution Property:**

   If $u \to 0$ as $x \to 0$, then $\lim ¿_{x \to 0} \frac{\sin u}{u} = 1¿$.

4. **Limit Product & Quotient Rules:**

$$\lim_{x \to a} [f(x) \cdot g(x)] = \left(\lim_{x \to a} f(x)\right) \cdot \left(\lim_{x \to a} g(x)\right)$$

Exercise 21

$$\lim_{\theta \to 0} \frac{\sin \sqrt{2\theta}}{\sqrt{2\theta}}$$

- **Step 1:** Let $u = \sqrt{2\theta}$.
- **Step 2:** As $\theta \to 0$, $u \to 0$.
- **Step 3:** Substitute u into the limit:

$$\lim_{u \to 0} \frac{\sin u}{u} = 1$$

Answer: 1

Exercise 22

$$\frac{\lim_{t \to 0} \sin k t}{t} \left(k \text{ is a constant} \right)$$

- **Step 1:** Multiply the numerator and denominator by k:

$$\lim_{t \to 0} \left(k \cdot \frac{\sin k t}{k t} \right)$$

- **Step 2:** Pull the constant k out of the limit:

$$k \cdot \frac{\lim_{t \to 0} \sin k t}{k t}$$

- **Step 3:** Let $u = k t$. As $t \to 0$, $u \to 0$:

$$k \cdot \left(\frac{\lim_{u \to 0} \sin u}{u} \right) = k \cdot (1) = k$$

Answer: k

Exercise 23

$$\frac{\lim_{y \to 0} \sin 3 y}{4 y}$$

- **Step 1:** Rewrite the denominator as $\frac{4}{3} \cdot (3 y)$:

$$\lim_{y \to 0} \left(\frac{3}{4} \cdot \frac{\sin 3 y}{3 y} \right)$$

- **Step 2:** Pull out the constant factor $\frac{3}{4}$:

$$\frac{3}{4} \cdot \frac{\lim_{y \to 0} \sin 3 y}{3 y}$$

- **Step 3:** Since $\lim ¿_{y \to 0} \frac{\sin 3 y}{3 y} = 1 ¿$:

$$\frac{3}{4}\cdot 1=\frac{3}{4}$$

Answer: $\frac{3}{4}$

Exercise 24

$$\frac{\lim\limits_{h\to 0^-} h}{\sin 3h}$$

- **Step 1:** Multiply numerator and denominator by 3:

$$\lim_{h\to 0^-}\left(\frac{1}{3}\cdot\frac{3h}{\sin 3h}\right)$$

- **Step 2:** Pull out $\frac{1}{3}$:

$$\frac{1}{3}\cdot\frac{\lim\limits_{h\to 0^-} 3h}{\sin 3h}$$

- **Step 3:** Since $\lim_{u\to 0}\frac{u}{\sin u}=1$:

$$\frac{1}{3}\cdot 1=\frac{1}{3}$$

Answer: $\frac{1}{3}$

Exercise 25

$$\frac{\lim\limits_{x\to 0}\tan 2x}{x}$$

- **Step 1:** Use identity $\tan 2x=\frac{\sin 2x}{\cos 2x}$:

$$\frac{\lim\limits_{x\to 0}\sin 2x}{x\cos 2x}$$

- **Step 2:** Rearrange terms:

$$\lim_{x \to 0} \left(2 \cdot \frac{\sin 2x}{2x} \cdot \frac{1}{\cos 2x} \right)$$

- **Step 3:** Evaluate limits individually:

$$2 \cdot \left(\frac{\lim_{x \to 0} \sin 2x}{2x} \right) \cdot \left(\frac{\lim_{x \to 0} 1}{\cos 2x} \right) = 2 \cdot (1) \cdot \left(\frac{1}{\cos 0} \right) = 2 \cdot 1 \cdot 1 = 2$$

Answer: 2

Exercise 26

$$\frac{\lim_{t \to 0} 2t}{\tan t}$$

- **Step 1:** Use identity $\tan t = \frac{\sin t}{\cos t}$:

$$\frac{\lim_{t \to 0} 2t}{\frac{\sin t}{\cos t}} = \lim_{t \to 0} \left(2 \cdot \frac{t}{\sin t} \cdot \cos t \right)$$

- **Step 2:** Apply limit laws:

$$2 \cdot \left(\frac{\lim_{t \to 0} t}{\sin t} \right) \cdot \left(\lim_{t \to 0} \cos t \right) = 2 \cdot (1) \cdot (1) = 2$$

Answer: 2

Exercise 27

$$\frac{\lim_{x \to 0} x \csc 2x}{\cos 5x}$$

- **Step 1:** Express $\csc 2x = \frac{1}{\sin 2x}$:

$$\frac{\lim_{x \to 0} x}{\sin 2x \cdot \cos 5x}$$

- **Step 2:** Multiply and divide by 2:

$$\lim_{x \to 0} \left(\frac{1}{2} \cdot \frac{2x}{\sin 2x} \cdot \frac{1}{\cos 5x} \right)$$

- **Step 3:** Evaluate limits as $x \to 0$:

$$\frac{1}{2} \cdot (1) \cdot \frac{1}{\cos 0} = \frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$$

Answer: $\frac{1}{2}$

Exercise 28

$$\lim_{x \to 0} 6x^2 (\cot x)(csc\, 2x)$$

- **Step 1:** Express in terms of sines and cosines:

$$\cot x = \frac{\cos x}{\sin x}, csc\, 2x = \frac{1}{\sin 2x}$$

- **Step 2:** Substitute trigonometric functions:

$$\lim_{x \to 0} \left(6x^2 \cdot \frac{\cos x}{\sin x} \cdot \frac{1}{\sin 2x} \right)$$

- **Step 3:** Rearrange into standard limit forms:

$$\lim_{x \to 0} \left(6 \cdot \frac{x}{\sin x} \cdot \frac{x}{\sin 2x} \cdot \cos x \right) = \lim_{x \to 0} \left(6 \cdot \frac{x}{\sin x} \cdot \frac{1}{2} \cdot \frac{2x}{\sin 2x} \cdot \cos x \right)$$

- **Step 4:** Simplify and evaluate:

$$3 \cdot \left(\frac{\lim_{x \to 0} x}{\sin x} \right) \cdot \left(\frac{\lim_{x \to 0} 2x}{\sin 2x} \right) \cdot \left(\lim_{x \to 0} \cos x \right) = 3 \cdot (1) \cdot (1) \cdot (1) = 3$$

Answer: 3

Exercise 29

$$\frac{\lim_{x \to 0} x + x \cos x}{\sin x \cos x}$$

- **Step 1:** Factor out x from the numerator:

$$\frac{\lim_{x \to 0} x(1+\cos x)}{\sin x \cos x}$$

- **Step 2:** Separate into two fractions:

$$\lim_{x \to 0} \left(\frac{x}{\sin x} \cdot \frac{1+\cos x}{\cos x} \right)$$

- **Step 3:** Evaluate each factor as $x \to 0$:

$$\left(\frac{\lim_{x \to 0} x}{\sin x} \right) \cdot \left(\frac{\lim_{x \to 0} 1+\cos x}{\cos x} \right) = (1) \cdot \left(\frac{1+1}{1} \right) = 1 \cdot 2 = 2$$

Answer: 2

Exercise 30

$$\frac{\lim_{x \to 0} x^2 - x + \sin x}{2x}$$

- **Step 1:** Divide each term in the numerator by $2x$:

$$\lim_{x \to 0} \left(\frac{x^2}{2x} - \frac{x}{2x} + \frac{\sin x}{2x} \right) = \lim_{x \to 0} \left(\frac{x}{2} - \frac{1}{2} + \frac{1}{2} \cdot \frac{\sin x}{x} \right)$$

- **Step 2:** Take the limit as $x \to 0$:

$$0 - \frac{1}{2} + \frac{1}{2}(1) = -\frac{1}{2} + \frac{1}{2} = 0$$

Answer: 0

Exercise 31

$$\frac{\lim_{t \to 0} \sin(1 - \cos t)}{1 - \cos t}$$

- **Step 1:** Let $u = 1 - \cos t$.
- **Step 2:** As $t \to 0$, $\cos t \to 1$, so $u = 1 - 1 \to 0$.
- **Step 3:** Transform the limit:

$$\frac{\lim_{u \to 0} \sin u}{u} = 1$$

Answer: 1

Exercise 32

$$\frac{\lim_{h \to 0} \sin(\sin h)}{\sin h}$$

- **Step 1:** Let $u = \sin h$.
- **Step 2:** As $h \to 0$, $u = \sin h \to 0$.
- **Step 3:** Transform the limit:

$$\frac{\lim_{u \to 0} \sin u}{u} = 1$$

Answer: 1

Exercise 33

$$\frac{\lim_{\theta \to 0} \sin \theta}{\sin 2\theta}$$

- **Step 1:** Divide numerator and denominator by θ:

$$\frac{\lim_{\theta \to 0} \frac{\sin \theta}{\theta}}{\frac{\sin 2\theta}{\theta}}$$

- **Step 2:** Multiply the denominator by $\frac{2}{2}$:

$$\frac{\lim_{\theta \to 0} \frac{\sin \theta}{\theta}}{2 \cdot \frac{\sin 2\theta}{2\theta}}$$

- **Step 3:** Evaluate limits as $\theta \to 0$:

$$\frac{1}{2\cdot(1)}=\frac{1}{2}$$

Answer: $\frac{1}{2}$

Exercise 34

$$\frac{\lim\limits_{x\to 0}\sin 5x}{\sin 4x}$$

- **Step 1:** Divide numerator and denominator by x:

$$\frac{\lim\limits_{x\to 0}\frac{\sin 5x}{x}}{\frac{\sin 4x}{x}}$$

- **Step 2:** Multiply numerator by $\frac{5}{5}$ and denominator by $\frac{4}{4}$:

$$\frac{\lim\limits_{x\to 0}5\cdot\frac{\sin 5x}{5x}}{4\cdot\frac{\sin 4x}{4x}}$$

- **Step 3:** Evaluate limits as $x\to 0$:

$$\frac{5\cdot(1)}{4\cdot(1)}=\frac{5}{4}$$

Answer: $\frac{5}{4}$

Exercise 35

$$\frac{\lim\limits_{x\to 0}\tan 3x}{\sin 8x}$$

- **Step 1:** Express $\tan 3x=\frac{\sin 3x}{\cos 3x}$:

$$\lim_{x \to 0} \left(\frac{\sin 3x}{\sin 8x} \cdot \frac{1}{\cos 3x} \right)$$

- **Step 2:** Divide $\sin 3x$ and $\sin 8x$ by x:

$$\lim_{x \to 0} \left(\frac{3 \cdot \frac{\sin 3x}{3x}}{8 \cdot \frac{\sin 8x}{8x}} \cdot \frac{1}{\cos 3x} \right)$$

- **Step 3:** Evaluate limits as $x \to 0$:

$$\frac{3 \cdot (1)}{8 \cdot (1)} \cdot \frac{1}{\cos 0} = \frac{3}{8} \cdot 1 = \frac{3}{8}$$

$$\text{Answer: } \frac{3}{8}$$

Exercise 36

$$\lim_{y \to 0} \frac{\sin 3y \cot 5y}{y \cot 4y}$$

- **Step 1:** Convert cotangents to sines and cosines ($\cot \theta = \frac{\cos \theta}{\sin \theta}$):

$$\frac{\sin 3y \cot 5y}{y \cot 4y} = \frac{\sin 3y \cdot \frac{\cos 5y}{\sin 5y}}{y \cdot \frac{\cos 4y}{\sin 4y}}$$

- **Step 2:** Rearrange into standard factors:

$$\left(\frac{\sin 3y}{y} \right) \cdot \left(\frac{\sin 4y}{\sin 5y} \right) \cdot \left(\frac{\cos 5y}{\cos 4y} \right)$$

- **Step 3:** Rewrite $\frac{\sin 4y}{\sin 5y}$ using standard form:

$$\frac{\sin 4y}{\sin 5y} = \frac{4 \cdot \frac{\sin 4y}{4y}}{5 \cdot \frac{\sin 5y}{5y}}$$

- **Step 4:** Evaluate each product as $y \to 0$:

- $\lim ¿_{y \to 0} \frac{\sin 3y}{y} = 3 \cdot (1) = 3 ¿$
- $\lim ¿_{y \to 0} \frac{\sin 4y}{\sin 5y} = \frac{4 \cdot (1)}{5 \cdot (1)} = \frac{4}{5} ¿$
- $\lim ¿_{y \to 0} \frac{\cos 5y}{\cos 4y} = \frac{\cos 0}{\cos 0} = \frac{1}{1} = 1 ¿$

- **Step 5:** Combine results:

$$3 \times \frac{4}{5} \times 1 = \frac{12}{5}$$

Answer: $\frac{12}{5}$